

Reamy Womodowe Achirobe

Mechanical Engineer · Thermofluids, CFD & Computational AI

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RESEARCH PROFILE

Mechanical engineer specialising in AI-driven thermofluids diagnostics, convolutional neural network design, and physics-informed modelling for energy systems. MPhil research produced SDCNN, a custom deep-learning framework for automated photovoltaic fault detection via infrared thermography, achieving a perfect diagnosis F1-score under five-fold cross-validation and outperforming three pre-trained baselines. Experimental background spans infrared thermography, fluidised-bed multiphase diagnostics (DAAD-funded, FAU Erlangen-Nuremberg), and CFD pedagogy (ANSYS Fluent). Currently teaching CFD and Fluid Mechanics at Ashesi University and pursuing research in digital twins and physics-informed surrogate modelling.

EDUCATION

MPhil in Mechanical Engineering (Thermofluids and Energy Systems) **2023 to 2025**
Kwame Nkrumah University of Science and Technology (KNUST), Ghana

- Awarded: Highest Cumulative Weighted Average, Department of Mechanical Engineering
- Funded: KEEP Scholarship (full funding)
- Thesis: *Using Machine Learning and Infrared Thermography to Identify and Classify Defects in Crystalline Photovoltaics in Ghana*
- Key coursework: Computational Fluid Dynamics, Advanced Engineering Thermodynamics, Fluid Machinery, Heat and Mass Transfer, Computer-Aided Analysis and Modelling, Dynamics

BSc in Mechanical Engineering (Second Class Upper Division) **2016 to 2020**
Kwame Nkrumah University of Science and Technology (KNUST), Ghana

- Key coursework: Fluid Mechanics I & II, Hydraulics & Pneumatics, Numerical Analysis, Renewable Energy Conversion, Strength of Materials, Probability & Statistics

PROFESSIONAL TRAINING

Marine New Energy & Advanced Manufacturing Technologies **Jul 2026**
Ocean University of China (OUC), Qingdao, China, 18-day intensive programme

- Marine renewable energy technologies, offshore wind blade manufacturing (CRRC New Era), deep-sea engineering (CIMC Raffles), and marine resource development (CNOOC)
- Classroom lectures, laboratory sessions, and industry visits at offshore energy facilities

PUBLICATIONS AND MANUSCRIPTS

Journal Articles

- Achirobe, R.W., Adaramola, M.S., and Mensah, L.D. (2026). SDCNN: A Shallow Dense Convolutional Neural Network for Automated Fault Detection and Diagnosis in Photovoltaic Systems via Infrared Thermography. *Engineering Applications of Artificial Intelligence*. [Under review]
- Achirobe, R.W. and Osei, D.A. (2026). Computer Vision Centric Digital Twins for Intelligent

Mobility: Simulation of Object Detection and Scene Segmentation. *International Journal of Innovative Research in Computer Technology (IJIRCT)*. [Accepted]

- Achirobe, R.W. et al. (2026). A physics-based digital twin for photovoltaic hotspot degradation coupling thermal conduction with Arrhenius kinetics. [In preparation]

Conference Papers

- Afari, A.A., Fiagbe, Y.A.K., Ramde, E.W., Alhassan, S.T., Quansah, E., and Achirobe, R.W. (2026). Dual Axis Solar Tracker with Low Operating Power Consumption. *International Conference on Sustainable Systems and Infrastructure Development (ICSSID 2026)*, Paper ID: 188. [Submitted]

RESEARCH AND TEACHING EXPERIENCE

Faculty Intern, CFD and Fluid Mechanics

Jan 2026 to Present

Ashesi University, Berekuso, Ghana

- Designed and deliver CFD (ME 431) and Fluid Mechanics (ME 432) courses end-to-end using ANSYS Fluent 2025 R2, covering external aerodynamics, converging-diverging nozzle flow, and airfoil analysis
- Implemented $k-\omega$ SST and $k-\epsilon$ turbulence models in laboratory exercises; authored simulation guides, problem sets, and examination papers
- Brought students from zero simulation experience to completing multi-physics ANSYS analyses; 100% course completion rate

Research Consultant

2025 to Present

Klerconsult, Ghana

- Applies data-driven modelling and computational methods to energy systems and infrastructure projects

Graduate Researcher, ML and Digital Twins for Energy Systems

2023 to 2025

KNUST, Kumasi, Ghana

- Designed SDCNN, a custom convolutional neural network trained from scratch (Python/TensorFlow) for PV fault detection via IRT; achieved mean detection F1-score of 0.884 and perfect diagnosis F1-score of 1.000 under five-fold cross-validation at Ghana's 2.5 MW Navrongo VRA Solar Plant
- Benchmarked against VGG-16, MobileNetV2, and EfficientNet-B0 with paired t-tests, Wilcoxon signed-rank tests, and Cohen's d; implemented Grad-CAM spatial explainability
- Built physics-based digital twin coupling thermal conduction with Arrhenius degradation kinetics for hotspot prediction (manuscript in preparation)
- Collaboration with Prof. M.S. Adaramola (NMBU, Norway) and the Energy Commission of Ghana

Research Assistant, DAAD-Funded Multiphase Flow Project

Sep to Nov 2024

Chair for Energy Process Engineering, Friedrich-Alexander-Universitat Erlangen-Nurnberg (FAU), Germany

- Characterised fluidised-bed hydrodynamics using dual-side pressure transducers, high-speed imaging, and optical diagnostics
- Applied FFT-based signal processing (Python/NumPy) and image analysis (FIJI) for bubble formation and particle agglomeration studies
- Developed and validated predictor schemes for particle agglomeration; measurement protocol adopted by the research group

- Led installation, commissioning, and maintenance of centrifugal pumps and pipeline systems across multiple client sites

TECHNICAL SKILLS

- **Programming and ML:** Python (TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, SciPy), MATLAB; CNNs, physics-informed ML, surrogate modelling, data pipelines
- **CFD and Simulation:** ANSYS Fluent, SpaceClaim, Discovery; turbulence modelling ($k-\omega$ SST, $k-\epsilon$); governing-equation modelling; physics-based digital twins
- **Experimental Methods:** Infrared thermography (FLIR), pressure transducer instrumentation, high-speed imaging, FFT-based signal processing, optical diagnostics (FIJI)
- **Tools:** L^AT_EX, SQL, Excel, Git/GitHub; Apple M1 with TensorFlow-Metal development environment

HONOURS AND AWARDS

- Highest Cumulative Weighted Average, Department of Mechanical Engineering, KNUST 2025
- KEEP Scholarship, Full funding for MPhil studies, KNUST 2023 to 2025
- DAAD Research Internship Grant, Friedrich-Alexander-Universitat Erlangen-Nurnberg, Germany 2024

LANGUAGES

English (Fluent, sole medium of instruction throughout all university studies) · Kasem (Native) · Twi (Fluent)

PROFESSIONAL AFFILIATIONS

- Vice President, IAESTE KNUST (2024 to 2025), coordinated international engineering trainee placements
- Student Member, Ghana Institution of Engineers (2025)
- Student Member, National Society of Black Engineers (2025)

REFERENCES

Prof. Muiyiwa S. Adaramola Professor, Faculty of Environmental Sciences Norwegian University of Life Sciences (NMBU) As, Norway MPhil co-supervisor and corresponding author muyiwa.adaramola@nmbu.no	Dr. Lena Dzifa Mensah Associate Professor, Dept. of Mechanical Engineering KNUST, Kumasi, Ghana Principal MPhil supervisor ldmensah.coe@knust.edu.gh
Dr. Heather Beem , Assistant Professor, Dept. of Engineering, Ashesi University, Ghana CFD teaching mentor and fluid mechanics pedagogy advisor · hbeem@ashesi.edu.gh	